

指数型② $a_{n+1} = p a_n + q \cdot p^n$

両辺 p^{n+1} でわって

$$\frac{a_{n+1}}{p^{n+1}} = \frac{a_n}{p^n} + \frac{q}{p}$$

と変形して□等差型へ

例B $a_{n+1} = 2a_n + 3 \cdot 2^n \dots \textcircled{1}, a_1 = 1$

(解法1)

①の両辺を 2^{n+1} でわって

$$\frac{a_{n+1}}{2^{n+1}} = \frac{a_n}{2^n} + \frac{3}{2}$$

$$\frac{a_n}{2^n} = b_n \text{ とおくと } \leftarrow \frac{a_{n+1}}{2^{n+1}} = b_{n+1}$$

$$\begin{cases} b_{n+1} = b_n + \frac{3}{2} \\ b_1 = \frac{a_1}{2^1} = \frac{1}{2} \end{cases} \quad \leftarrow \square \text{等差型に帰着}$$

よって

$$\begin{aligned} b_n &= \frac{1}{2} + (n-1) \cdot \frac{3}{2} \\ &= \frac{1}{2} (3n-2) \end{aligned}$$

ここで, $\frac{a_n}{2^n} = b_n$ より

$$\begin{aligned} a_n &= 2^n \cdot b_n \\ &= 2^n \cdot \frac{1}{2} (3n-2) \\ &= 2^{n-1} (3n-2), \end{aligned}$$

例B $a_{n+1} = 2a_n + 3 \cdot 2^n \dots \textcircled{1}, a_1 = 1$

(解法2)

$$\boxed{a_{n+1} + A(n+1)2^{n+1}} = 2 \boxed{a_n + A \cdot 2^n} \quad \leftarrow \text{幸せな形}$$

$n \in n+1$ にした式 指数部

$$\begin{aligned} a_{n+1} + (A+1)2^{n+1} &= 2a_n + 2A \cdot 2^{n+1} \\ a_{n+1} &= 2a_n - A \cdot 2^{n+1} \end{aligned}$$

①と係数比較して

$$A = -\frac{3}{2}$$

$$a_{n+1} - 3(n+1)2^n = 2(a_n - 3n \cdot 2^{n-1})$$

$$\left(\{a_n - 3n \cdot 2^{n-1}\} \text{ は } \textcircled{2} a_1 - 3 \cdot 1 = \textcircled{3} 2 \text{ の等比数列} \right)$$

$$\begin{aligned} a_n - 3n \cdot 2^{n-1} &= (a_1 - 3 \cdot 1) \cdot 2^{n-1} \\ &= -2 \cdot 2^{n-1} \end{aligned}$$

$$\therefore a_n = 2^{n-1} (3n-2),$$